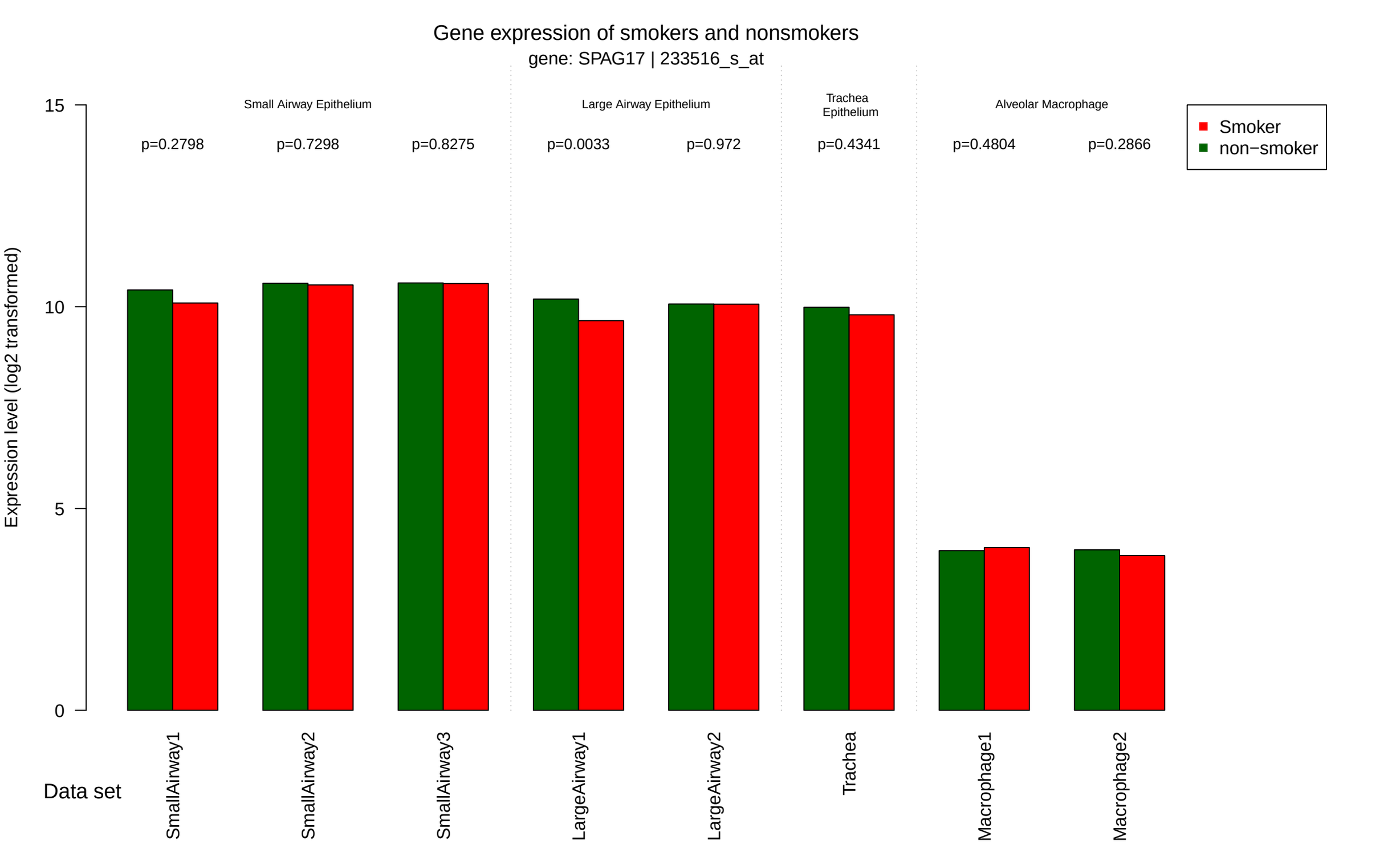
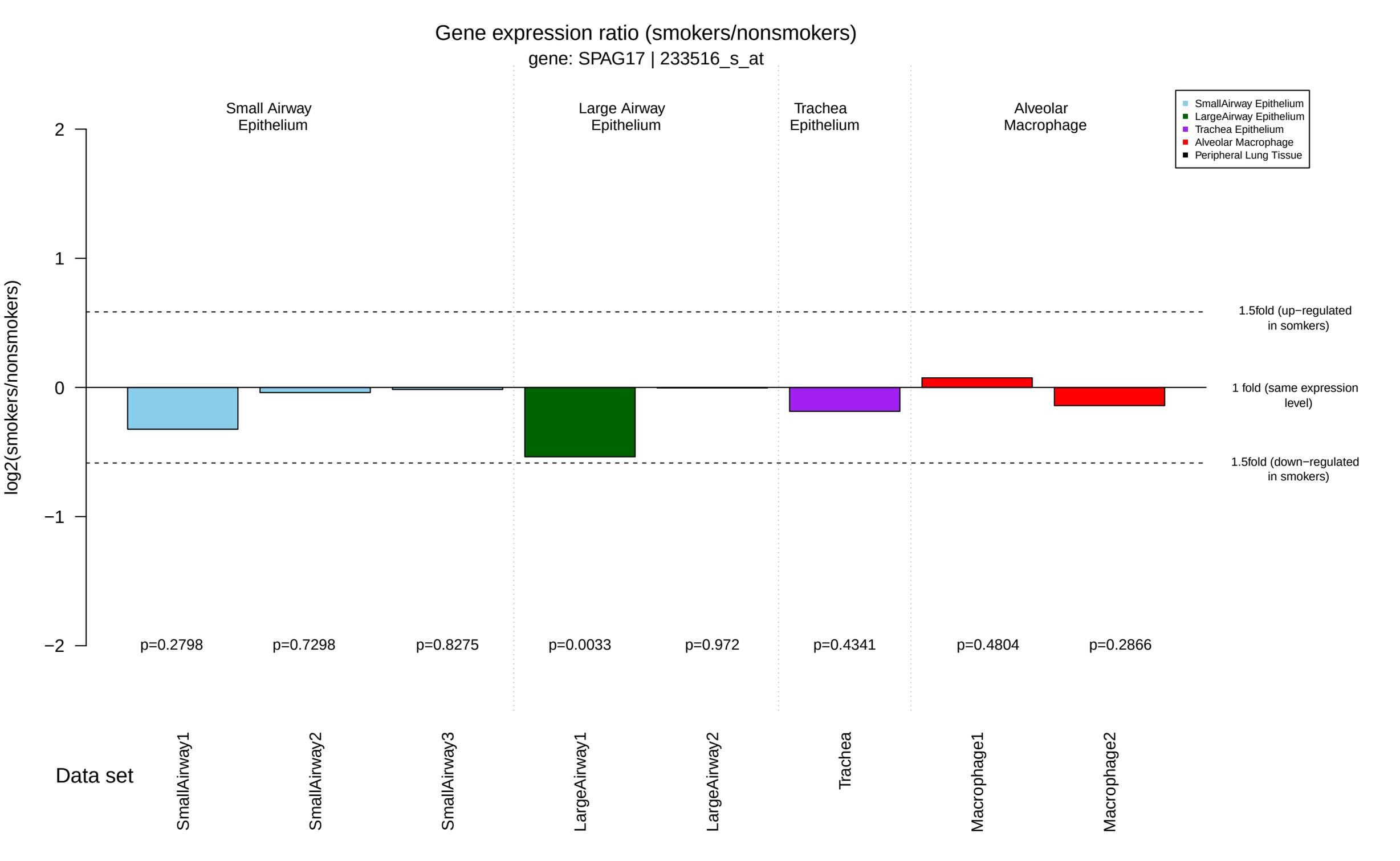
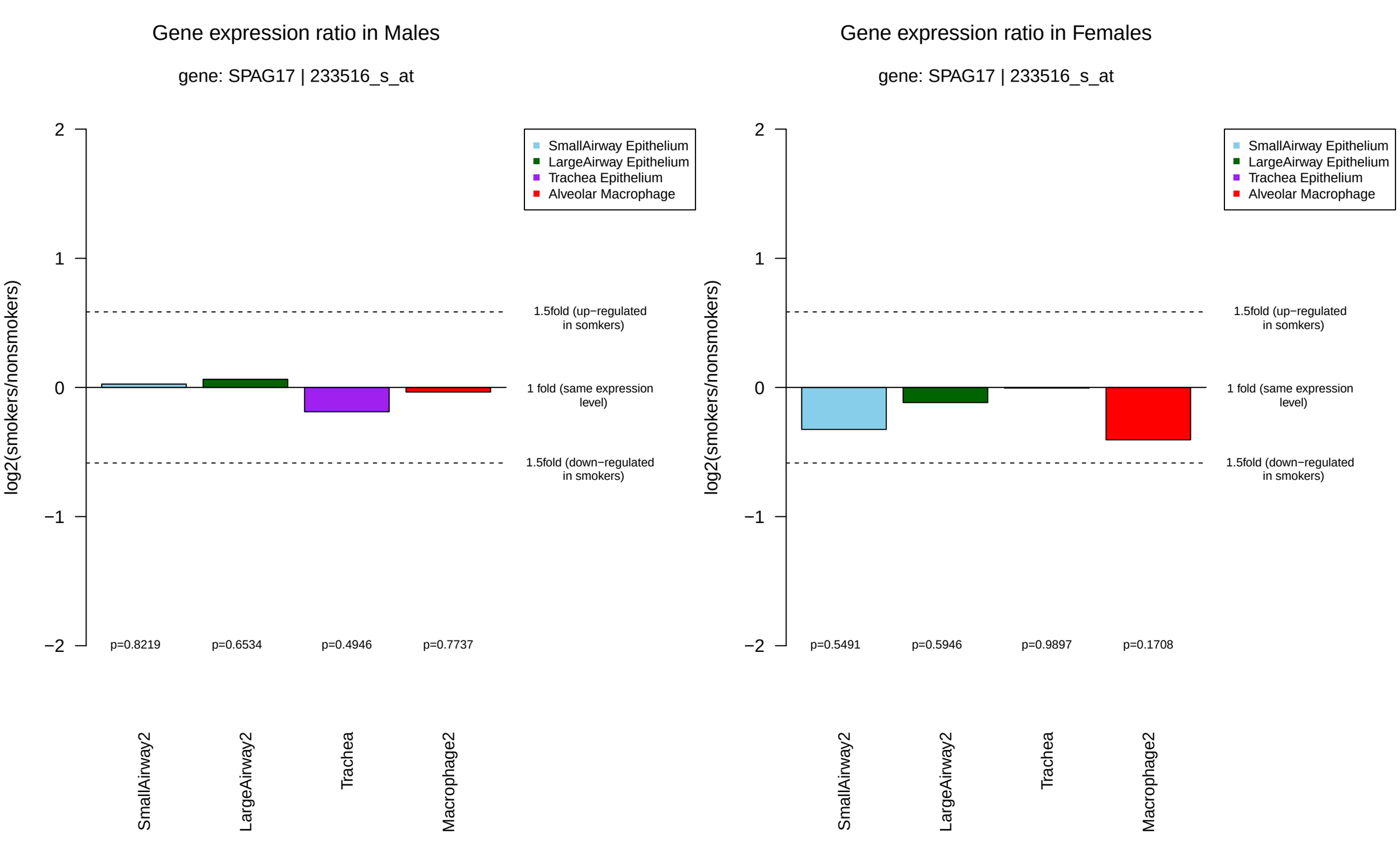
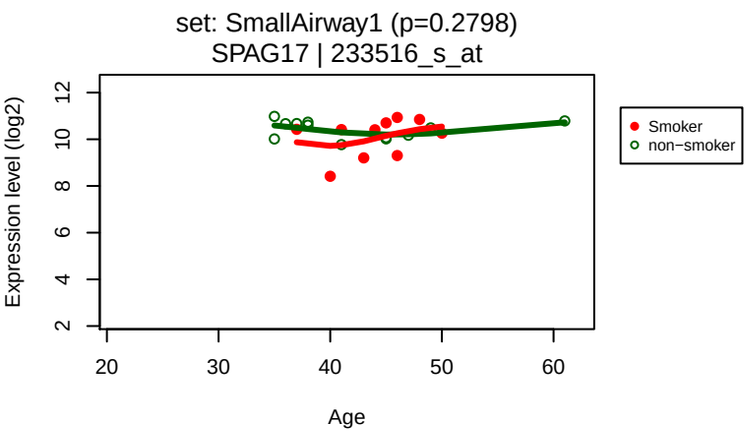
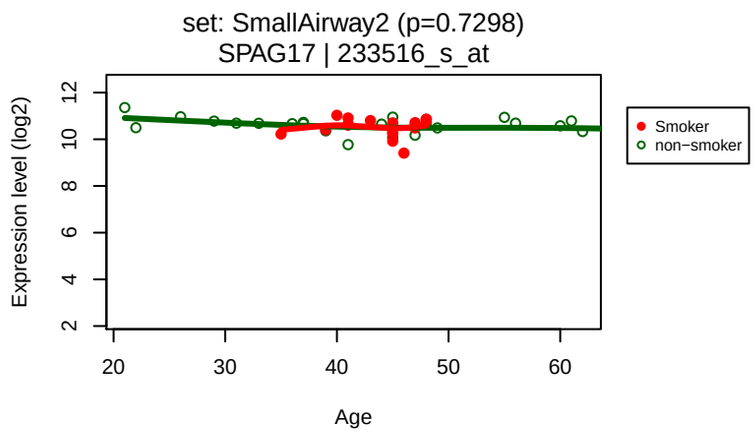
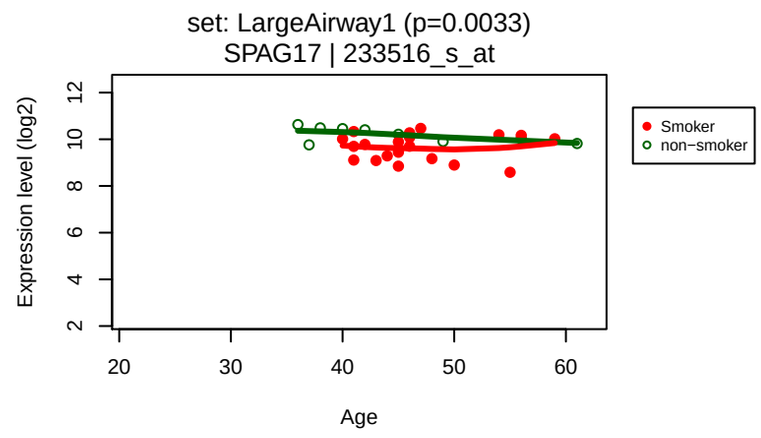

Gene expression of smokers and nonsmokers
gene: SPAG17 | 233516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2798
p=0.7298
p=0.8275
p=0.0033
p=0.972
p=0.4341
p=0.4804
p=0.2866
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

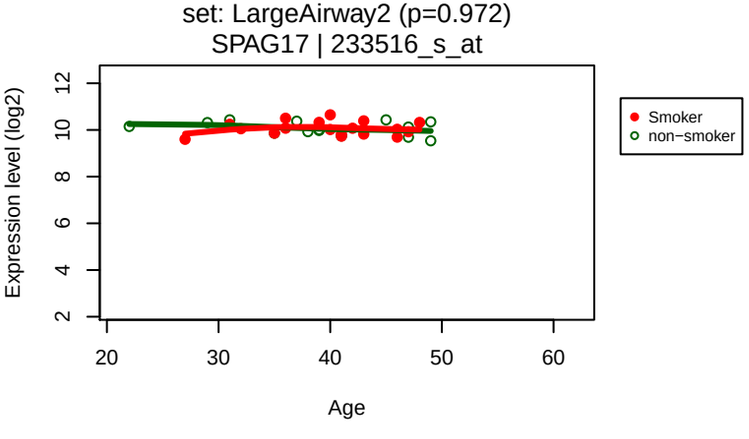

Gene expression ratio (smokers/nonsmokers)
gene: SPAG17 | 233516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2798
p=0.7298
p=0.8275
p=0.0033
p=0.972
p=0.4341
p=0.4804
p=0.2866
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

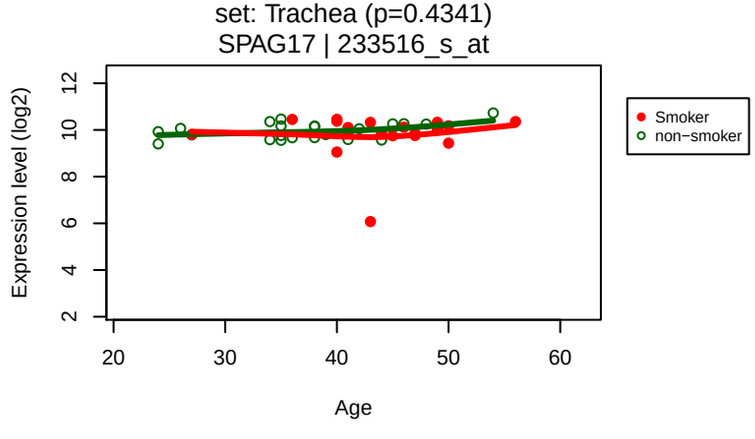

Gene expression ratio in Males
gene: SPAG17 | 233516_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8219
p=0.6534
p=0.4946
p=0.7737
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SPAG17 | 233516_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5491
p=0.5946
p=0.9897
p=0.1708
SmallAirway2
LargeAirway2
Trachea
Macrophage2

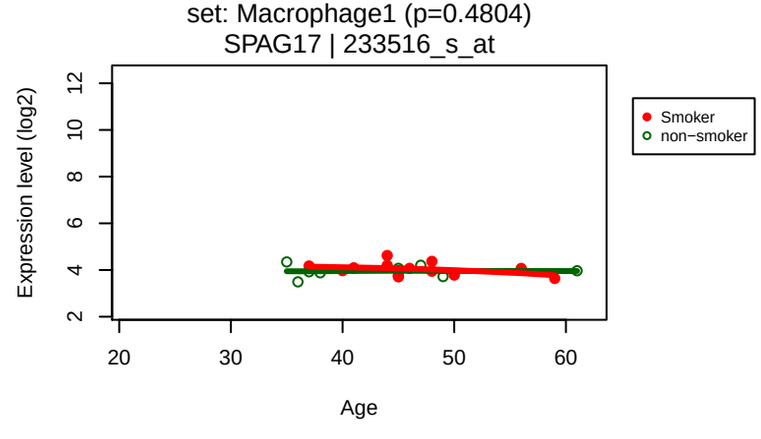

set: SmallAirway1 (p=0.2798)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker

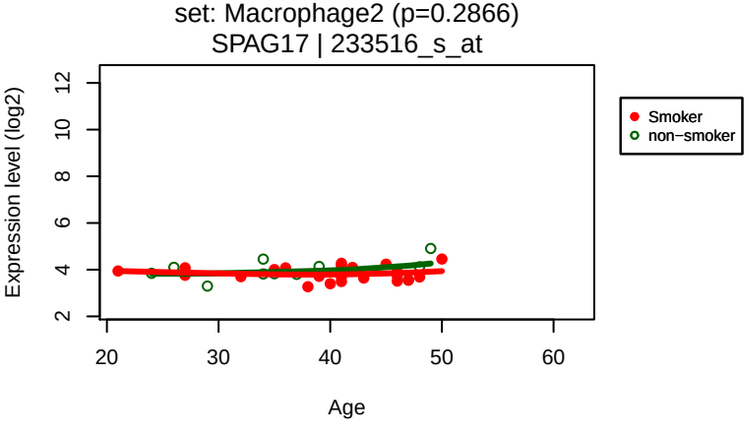

set: SmallAirway2 (p=0.7298)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker

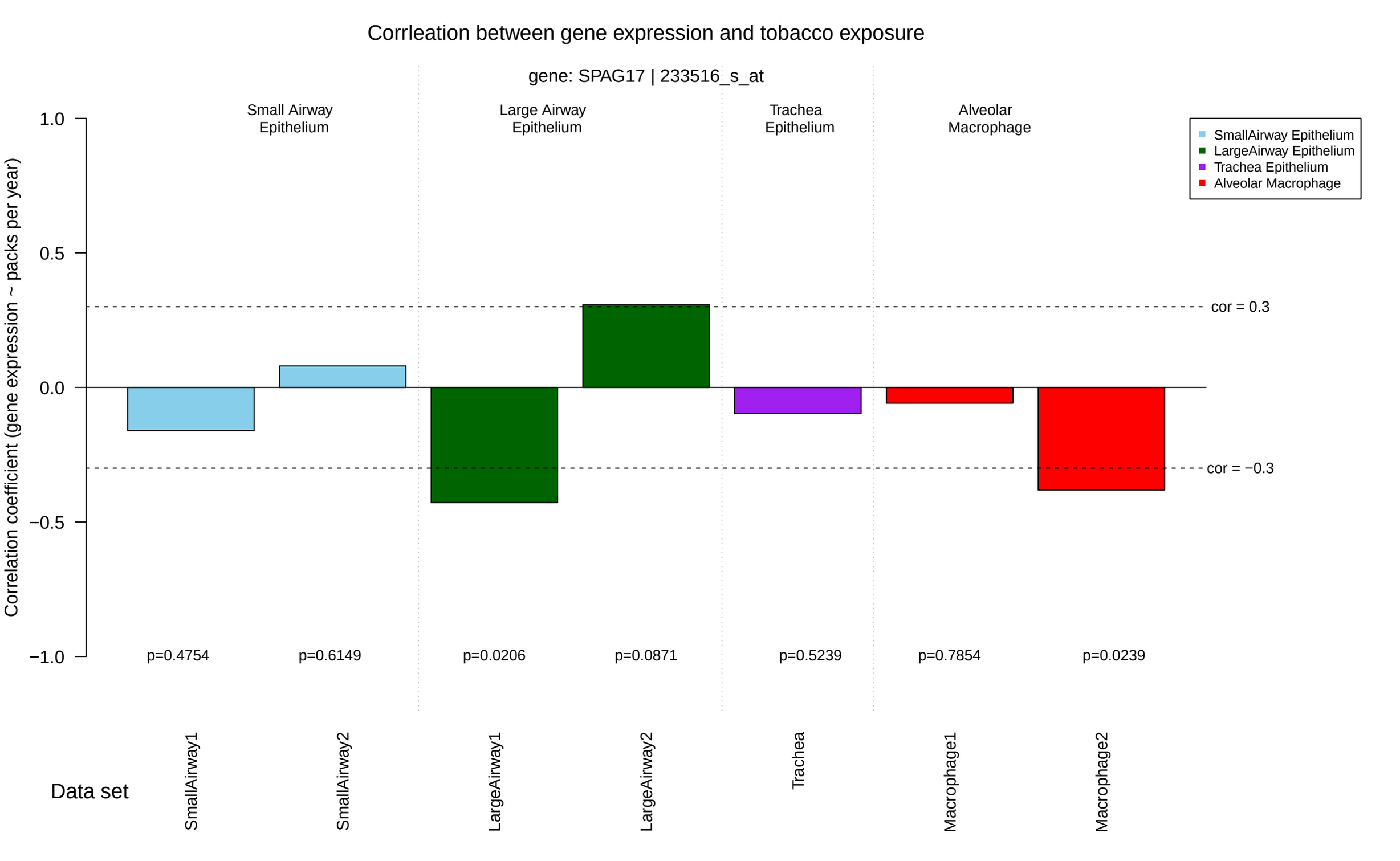

set: LargeAirway1 (p=0.0033)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.972)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4341)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4804)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2866)
SPAG17 | 233516_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SPAG17 | 233516_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4754
p=0.6149
p=0.0206
p=0.0871
p=0.5239
p=0.7854
p=0.0239
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2